Group B Topic-Specific Data Extraction

15 report forms | 45 reported outcome rows | 2 separately calculated outcomes

Report forms are not final included studies. Units, conditions, baselines and source anchors control every interpretation. Not extracted means unchecked; not reported is limited to the checked section stated in the cell. Source preprints, cluster-only results, training results and non-GPU results are not silently converted into eligible local inference evidence.

Report	Study linkage	Title	Eligibility
GB-P001	pending_study_linkage	VastPipe	provisionally_eligible
GB-P002	pending_study_linkage	iGniter	provisionally_eligible
GB-P003	pending_study_linkage	gpulets	provisionally_eligible
GB-P004	pending_study_linkage	PipeSwitch	provisionally_eligible
GB-P005	pending_study_linkage	REEF conference	eligible_companion_candidate
GB-P006	GB-PS006	Paella: Low-latency Model Serving with Software-defined GPU Scheduling	provisionally_eligible;study_identity_provisional
GB-P007	GB-PS007	Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications	provisionally_eligible;study_identity_provisional
GB-P008	GB-PS008	GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform	provisionally_eligible;study_identity_provisional
GB-P009	GB-PS009	Serving DNNs like Clockwork: Performance Predictability from the Bottom Up	provisionally_eligible;study_identity_provisional
GB-P010	GB-PS010	Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)	provisionally_eligible_AI_source_assessment
GB-P011	GB-PS011	ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments	uncertain_publication_version;technical_scope_supported
GB-P012	GB-PS012	Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference	provisionally_eligible_AI_source_assessment
GB-P013	GB-PS013	StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow	provisionally_eligible_AI_source_assessment
GB-P014	GB-PS014	Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)	provisionally_eligible_AI_source_assessment
GB-P015	GB-PS015	XSched: Preemptive Scheduling for Diverse XPUs	provisionally_eligible_AI_source_assessment

GB-P001 | VastPipe

Field	Registered evidence
authors	Li-Xian Ma and Le-Ping Wang and En Shao and Rong-Yu Cao and Guang-Ming Tan
year	2025
venue	Journal of Computer Science and Technology
doi	10.1007/s11390-024-3773-5
source	s11390-024-3773-5.md
source anchor	pp457-459, sections6.7-6.9, Figs14-16
mechanism	tuned/default MPS comparison; prior source check only
gpu hardware	NVIDIA A100 (existing source-check register)
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	not_extracted
models	VGG16+GoogLeNet (existing extraction rows)
tenant mix	not_extracted
batch load slo	not_extracted
baseline	No MPS / default MPS;exact quotas not extracted
latency	not extracted
throughput	not extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	not_extracted
tradeoffs	not_extracted

Reported outcomes

Outcome ID	GB-O001
Metric / value / unit	Throughput ratio | 2.7 | x
Hardware / mechanism	NVIDIA A100 | MPS tuned quotas
Workload / conditions	VGG16+GoogLeNet
Baseline	No MPS
Source anchor	pp457-458 Fig14 and accompanying prose
Limit / interpretation	Batch/rate/quotas unresolved; not tail-latency evidence
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O002
Metric / value / unit	Throughput ratio | 1.5 | x
Hardware / mechanism	NVIDIA A100 | MPS tuned quotas
Workload / conditions	VGG16+GoogLeNet
Baseline	Default MPS
Source anchor	pp457-458 Fig14 and accompanying prose
Limit / interpretation	Same comparison family; not a universal average
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P002 | iGniter

Field	Registered evidence
authors	Fei Xu and Jianian Xu and Jiabin Chen and Li Chen and Ruitao Shang and Zhi Zhou and Fangming Liu
year	2023
venue	IEEE Transactions on Parallel and Distributed Systems
doi	10.1109/TPDS.2022.3232715
source	iGniter user-supplied Markdown
source anchor	pp820-824, sections4.2,5.1-5.4, Figs13-17
mechanism	MPS quota allocation and shadow process recovery;prior source check
gpu hardware	NVIDIA V100 (existing source-check register)
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	not_extracted
models	AlexNet;ResNet50;VGG19;SSD
tenant mix	not_extracted
batch load slo	batch3 for prediction check;W1 SLO10ms for recovery example
baseline	Measured latency for prediction error;10ms SLO for recovery
latency	not extracted
throughput	not extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	not_extracted
tradeoffs	not_extracted

Reported outcomes

Outcome ID	GB-O003
Metric / value / unit	Prediction error range | 1.53-5.02 | percent
Hardware / mechanism	NVIDIA V100 | MPS 25% per model
Workload / conditions	AlexNet/ResNet50/VGG19/SSD; batch3
Baseline	Measured local latency
Source anchor	p822 section5.2 Fig13
Limit / interpretation	Prediction accuracy, not utilization or latency speedup
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O004
Metric / value / unit	Initial p99 latency | 15.6 | ms
Hardware / mechanism	NVIDIA V100 | MPS shadow-process recovery
Workload / conditions	W1 with W5 and W11
Baseline	W1 latency SLO10ms
Source anchor	p823 section5.3 Fig17 prose
Limit / interpretation	Transient violation precedes recovery; no unconditional guarantee
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P003 | gpulets

Field	Registered evidence
authors	Seungbeom Choi et al.
year	2022
venue	USENIX ATC22
doi	not resolved
source	https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf
source anchor	pp201-203 local mechanism/Fig7; p207 Tables2-3;pp208-209 system methodology
mechanism	MPS spatial partitions plus temporal batching; interference-aware gpulet scheduler
gpu hardware	RTX2080Ti11GB; two GPUs/server, two servers; local pair measurements separated
cpu memory	20-core Xeon E5-2630v4;192GB DRAM (Table2)
os kernel	Ubuntu18.04; kernel not extracted
driver runtime	NVIDIA440.64;CUDA10.2;PyTorch1.10 (p207 Table2)
models	GoogLeNet,LeNet,ResNet50,SSD-MobileNet,VGG16,MnasNet,MobileNetv2,DenseNet ,BaseBERT (Table3)
tenant mix	paired local inferences; system scenarios contain several models
batch load slo	Poisson inter-arrivals; model SLO5-202ms;99% requests within SLO; median of3 runs (pp207-208)
baseline	Solo latency for local Fig7; SBP and greedy best-fit for system scheduling
latency	Fig7 across-pair CDF threshold17.66%; not request-p90
throughput	not extracted
resource usage	not_extracted
scalability	8 logical backend servers with additional GPUs; system-level result (pp208-209)
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	20-second reorganization interval; overlap new proxy preparation with active serving (p207 Fig11)
advantages	not_extracted
limitations	Original Nexus not directly comparable; SBP ported; cross-model interference remains; no extracted security/fault test
tradeoffs	Partition and batch jointly selected; throughput numbers at system scale cannot be assigned to one GPU

Reported outcomes

Outcome ID	GB-O006
Metric / value / unit	Distribution threshold:90% of measured pairs | 17.66 | percent overhead below threshold
Hardware / mechanism	RTX2080Ti11GB | MPS spatial co-location
Workload / conditions	Paired model inferences
Baseline	Solo-running latency
Source anchor	p203 Fig7
Limit / interpretation	Across-pair CDF, not request-p90/p99; no hostile-memory/fault test
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P004 | PipeSwitch

Field	Registered evidence
authors	Zhihao Bai;Zhen Zhang;Yibo Zhu;Xin Jin
year	2020
venue	USENIX OSDI20
doi	not resolved
source	https://www.usenix.org/system/files/osdi20-bai.pdf
source anchor	p508 section6/6.1; p509 Tables3-4
mechanism	Temporal task switching; pipelined model transfer; active/standby workers and shared memory daemon
gpu hardware	V10016GB PCIe3x16;T416GB PCIe3x8
cpu memory	V100 instance:8vCPU Xeon E5-2686v4/61GB;T4 instance:8vCPU Platinum8259CL/32GB (p508)
os kernel	not_extracted
driver runtime	CUDA10.1;PyTorch1.3.0;torchvision0.4.2;scipy1.3.2;driver not extracted
models	ResNet152;Inception_v3;Bert_base
tenant mix	one training task preempted for one inference task; separable inference latency
batch load slo	training32;inference8;100-run averages;Fig6 reports minimum/maximum (p508)
baseline	Ready model;stop-and-start;MPS using unified memory
latency	Additional total latency versus ready model;Table3;not p99
throughput	not extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	Table3 total extra latency;Table4 first-layer startup overhead;kept separate
advantages	Reduced switching overhead for checked workloads; see extraction.csv
limitations	Mixed training/inference applicability;MPS benchmark swaps memory;modified PyTorch for all comparators;no extracted OS/driver/security/fault evidence
tradeoffs	Pipelined transfers and registered resident host models require runtime integration;inference-only multi-tenant relevance needs human eligibility check

Reported outcomes

Outcome ID	GB-O005
Metric / value / unit	Additional end-to-end latency | 6.01 | ms
Hardware / mechanism	NVIDIA V10016GB | Temporal switching
Workload / conditions	ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted; MPS overhead307.02ms
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Mean100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O007
Metric / value / unit	Additional end-to-end latency | 307.02 | ms
Hardware / mechanism	NVIDIA V10016GB | NVIDIA MPS with unified-memory swapping
Workload / conditions	ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O008
Metric / value / unit	Additional end-to-end latency | 5.40 | ms
Hardware / mechanism	NVIDIA V10016GB | Temporal switching
Workload / conditions	Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O009
Metric / value / unit	Additional end-to-end latency | 232.25 | ms
Hardware / mechanism	NVIDIA V10016GB | NVIDIA MPS with unified-memory swapping
Workload / conditions	Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O010
Metric / value / unit	Additional end-to-end latency | 10.27 | ms
Hardware / mechanism	NVIDIA V10016GB | Temporal switching
Workload / conditions	Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O011
Metric / value / unit	Additional end-to-end latency | 204.52 | ms
Hardware / mechanism	NVIDIA V10016GB | NVIDIA MPS with unified-memory swapping
Workload / conditions	Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O012
Metric / value / unit	Additional end-to-end latency | 5.57 | ms
Hardware / mechanism	NVIDIA T416GB | Temporal switching
Workload / conditions	ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O013
Metric / value / unit	Additional end-to-end latency | 259.20 | ms
Hardware / mechanism	NVIDIA T416GB | NVIDIA MPS with unified-memory swapping
Workload / conditions	ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O014
Metric / value / unit	Additional end-to-end latency | 7.66 | ms
Hardware / mechanism	NVIDIA T416GB | Temporal switching
Workload / conditions	Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O015
Metric / value / unit	Additional end-to-end latency | 193.05 | ms
Hardware / mechanism	NVIDIA T416GB | NVIDIA MPS with unified-memory swapping
Workload / conditions	Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O016
Metric / value / unit	Additional end-to-end latency | 34.56 | ms
Hardware / mechanism	NVIDIA T416GB | Temporal switching
Workload / conditions	Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O017
Metric / value / unit	Additional end-to-end latency | 338.25 | ms
Hardware / mechanism	NVIDIA T416GB | NVIDIA MPS with unified-memory swapping
Workload / conditions	Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Source anchor	p509 Table3; setup p508 section6
Limit / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P005 | REEF conference

Field	Registered evidence
authors	Mingcong Han;Hanze Zhang;Rong Chen;Haibo Chen
year	2022
venue	USENIX OSDI22
doi	not resolved
source	https://www.usenix.org/system/files/osdi22-han.pdf
source anchor	p548 section7.1 Table2;p549 section7.2 Fig11;section8 idempotence limitation
mechanism	Reset-based preemption;idempotent inference kernels;dynamic kernel padding;runtime/driver changes
gpu hardware	AMD Radeon Instinct MI50:60CUs/16GB;REEF-N extension NVIDIA V100
cpu memory	Intel Corei7-10700:8cores;16GB host DRAM (p548)
os kernel	Ubuntu18.04;kernel not extracted
driver runtime	ROCm4.3.0;ApacheTVM0.8.0;NVIDIA extension CUDA10.2
models	ResNet152;DenseNet201;VGG19;Inceptionv3;DistilBERT
tenant mix	DISB A/B:1RT+1BE;C:1RT+5BE;D/E:5RT+5BE
batch load slo	Table2 RT100/220/100/20/20req/s;D uniform,E Poisson;BE closed-loop;batch not extracted
baseline	RT-Only;SEQ;GPUStreams;separate wait-based preemption microbenchmark
latency	RT end-to-end overhead below2% in REAL;no request percentile extracted
throughput	RT+BE total normalized to RT-Only;not RT-only throughput
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	Checked workloads retain near RT-only latency while using idle capacity for BE inference
limitations	Reset requires idempotent kernels;model-pair padding incompatibilities;conference/journal report linkage open;not security/fault isolation
tradeoffs	DISB D/E throughput about25% below GPUStreams while limiting RT latency;implementation differs on closed-source GPUs

Reported outcomes

Outcome ID	GB-O018
Metric / value / unit	Overall throughput ratio | 1.60 | x
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	Includes RT and BE requests; RT-only reference dedicates GPU to RT tasks.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O019
Metric / value / unit	RT end-to-end latency overhead | 0.5 | percent
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	Described as negligible; not a p99 value or worst-case guarantee.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O020
Metric / value / unit	Overall throughput ratio | 1.14 | x
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	High RT load; differs from low-load DISB A.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O021
Metric / value / unit	RT end-to-end latency overhead | 1 | percent
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	About60 microseconds in prose; not a tail percentile.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O022
Metric / value / unit	RT end-to-end latency overhead | <1.5 | percent
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	DISB C-E:multiple BE clients
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	Authors also state0.1ms; keep inequality and scenario range.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O023
Metric / value / unit	Overall throughput ratio | 3.00 | x
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	DISB D:5 RT model clients20req/s uniform +5 BE model clients
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O024
Metric / value / unit	Overall throughput ratio | 2.96 | x
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	DISB E:5 RT model clients20req/s Poisson +5 BE model clients
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O025
Metric / value / unit	Overall throughput ratio | 7.7 | x
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	Apollo REAL trace about43 RTreq/s +5 BE model clients
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	Real-world trace mapped to representative DNN models; not production deployment measurement.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O026
Metric / value / unit	RT end-to-end latency overhead | <2 | percent
Hardware / mechanism	AMD MI5016GB | Reset-based kernel preemption + dynamic kernel padding
Workload / conditions	Apollo REAL trace about43 RTreq/s +5 BE model clients
Baseline	RT-Only dedicated GPU
Source anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limit / interpretation	Not a p99 latency claim or security/fault isolation test.
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P006 | Paella: Low-latency Model Serving with Software-defined GPU Scheduling

Field	Registered evidence
authors	Kelvin K.W. Ng; Henri Maxime Demoulin; Vincent Liu
year	2023
venue	SOSP 2023
doi	10.1145/3600006.3613163
source	https://kelvin-ng.github.io/assets/sosp2023-final224.pdf
source anchor	printed pp604-606; PDF pp10-12; sections7.1-7.3
mechanism	Instrumented kernels; software scheduling
gpu hardware	Tesla T4
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	Linux4.15.0;CUDA11.7;TVM0.10.0
models	Mixed inference; lognormal arrivals sigma1.5/2
tenant mix	Mixed inference; lognormal arrivals sigma1.5/2
batch load slo	not_extracted
baseline	CUDA-MS;Triton23.03;Paella ablations
latency	not_extracted
throughput	not_extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	Warmup excluded; TVM wrapper; MPS omitted from eight-model experiment
tradeoffs	Fairness versus short-job latency; instrumentation cost

Reported outcomes

Outcome ID	GB-O027
Metric / value / unit	ResNet18 p99 reduction factor | up to3 | x
Hardware / mechanism	Tesla T4 | Instrumented kernels; software scheduling
Workload / conditions	ResNet18+InceptionV3;lognormal sigma1.5/2
Baseline	CUDA-MS
Source anchor	printed p604 section7.1/Fig12; PDF pp10-11
Limit / interpretation	Author-reported approximate maximum; not uniform across loads
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O028
Metric / value / unit	p90 instrumentation overhead | 6.6 | microseconds
Hardware / mechanism	Tesla T4 | Instrumented kernels; software scheduling
Workload / conditions	Empty kernel;160 blocks;instrumentation with aggregation
Baseline	Uninstrumented kernel
Source anchor	printed p606 section7.3/Fig15; PDF p12
Limit / interpretation	Microbenchmark; not inference request latency
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P007 | Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

Field	Registered evidence
authors	Foteini Strati; Xianzhe Ma; Ana Klimovic
year	2024
venue	EuroSys 2024
doi	10.1145/3627703.3629578
source	https://anakli.inf.ethz.ch/papers/orion_eurosys24.pdf
source anchor	printed pp1082-1087; PDF pp8-13; sections6-7
mechanism	Interception; interference-aware kernel scheduling
gpu hardware	V10016GB;A10040GB extension
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	PyTorch1.12;Python3.9;CUDA10.2/11.3
models	High-priority plus best-effort inference;Poisson
tenant mix	High-priority plus best-effort inference;Poisson
batch load slo	not_extracted
baseline	Dedicated high-priority GPU;MPS;Streams;REEF-N policy
latency	not_extracted
throughput	not_extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	Trusted clients; no secure isolation; offline profiling; REEF-N reimplementation
tradeoffs	Tail latency versus best-effort progress

Reported outcomes

Outcome ID	GB-O029
Metric / value / unit	Aggregate throughput ratio | up to7.3 | x
Hardware / mechanism	V10016GB;A10040GB extension | Interception; interference-aware kernel scheduling
Workload / conditions	Two inference clients;Poisson;V100
Baseline	Dedicated high-priority inference
Source anchor	printed p1086 section6.2.3/Fig12; PDF p12
Limit / interpretation	Maximum; includes best-effort work
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O030
Metric / value / unit	High-priority p99 overhead | 15 | percent
Hardware / mechanism	V10016GB;A10040GB extension | Interception; interference-aware kernel scheduling
Workload / conditions	Two inference clients;Poisson;V100
Baseline	Dedicated high-priority inference
Source anchor	printed p1086 section6.2.3/Fig12; PDF p12
Limit / interpretation	Authors summarize within15%; averaged across collocations, not worst-case guarantee
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P008 | GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

Field	Registered evidence
authors	Aditya Dhakal; Sameer G Kulkarni; K.K. Ramakrishnan
year	2020
venue	SoCC 2020
doi	10.1145/3419111.3421284
source	https://par.nsf.gov/servlets/purl/10299299
source anchor	printed pp499-501; PDF pp8-10; sections4.1-4.3
mechanism	MPS allocation; adaptive batching
gpu hardware	One V10016GB
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	Linux4.4.0-150;DPDK19.02;CNTK/PyTorch/TensorRT
models	Concurrent inference functions;25ms target
tenant mix	Concurrent inference functions;25ms target
batch load slo	not_extracted
baseline	Default MPS;MPS+SLAB;TensorRT19.02
latency	not_extracted
throughput	not_extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	GUE is composite, not hardware utilization percentage; no extracted tail percentile
tradeoffs	Batching/fairness versus individual latency

Reported outcomes

Outcome ID	GB-O031
Metric / value / unit	GPU utilization efficacy GUE | 94.04 | composite score
Hardware / mechanism	One V10016GB | MPS allocation; adaptive batching
Workload / conditions	Alexnet+ResNet50;25ms target
Baseline	Default MPS GUE38.82
Source anchor	printed p500 Table3; PDF p9
Limit / interpretation	Not percent GPU utilization
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O032
Metric / value / unit	Signed latency improvement | -7.52 | percent
Hardware / mechanism	One V10016GB | MPS allocation; adaptive batching
Workload / conditions	ResNet50;batch1;GPU-ready batch execution
Baseline	TensorRT server19.02
Source anchor	printed p501 Table4; PDF p10
Limit / interpretation	Negative means worse latency; excludes transport overhead; single-model comparison
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P009 | Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

Field	Registered evidence
authors	Arpan Gujarati; Reza Karimi; Safya Alzayat; Wei Hao; Antoine Kaufmann; Ymir Vigfusson; Jonathan Mace
year	2020
venue	USENIX OSDI 2020
doi	not resolved
source	https://people.mpi-sws.org/~antoinek/documents/20osdi_clockwork.pdf
source anchor	printed pp451-452; author PDF pp9-10; section6.1/Fig5
mechanism	Predictable temporal scheduling
gpu hardware	One V10032GB in section6.1
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	Not extracted
models	15 ResNet50 instances;16 closed-loop clients/model
tenant mix	15 ResNet50 instances;16 closed-loop clients/model
batch load slo	not_extracted
baseline	Clipper;INFaaS (different EC2 hardware)
latency	not_extracted
throughput	not_extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	INFaaS comparison qualitative; fault tolerance unaddressed
tradeoffs	Predictability versus admission and concurrency

Reported outcomes

Outcome ID	GB-O033
Metric / value / unit	Clockwork goodput | approximately800 | requests/second
Hardware / mechanism	One V10032GB in section6.1 | Predictable temporal scheduling
Workload / conditions	15 ResNet50 instances;16 clients/model;500ms SLO
Baseline	INFaaS on nonidentical EC2 setup
Source anchor	printed p452 section6.1/Fig5; author PDF p10
Limit / interpretation	Within-SLO successes; comparator qualitative; not total throughput
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

Outcome ID	GB-O034
Metric / value / unit	Clockwork request latency | approximately300 | milliseconds
Hardware / mechanism	One V10032GB in section6.1 | Predictable temporal scheduling
Workload / conditions	15 ResNet50 instances;16 clients/model;500ms SLO
Baseline	No comparative ratio
Source anchor	printed p452 section6.1/Fig5; author PDF p10
Limit / interpretation	Batch16 scheduling explanation; not a p99 value
Assessor / status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending

GB-P010 | Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)

Field	Registered evidence
authors	Jiali Wang; Yankui Wang; Mingcong Han; Rong Chen
year	2025
venue	USENIX ATC2025, pp1657-1675
doi	not resolved
source	https://www.usenix.org/system/files/atc25-wang-jiali.pdf
source anchor	printed pp1665-1666, section6/6.1, Table3, Figure10; PDF pp10-11 (includes cover)
mechanism	hybrid spatial sharing; CUDA VMM memory handover and dynamic SM masking; separate inference/training runtimes
gpu hardware	single NVIDIA Tesla V100 16GB in section6.1; four-V100 host; separate section6.6 single A100 80GB LLM experiment
cpu memory	two Intel Xeon Gold6138, 80 total cores, 503GB DRAM; PCIe3.0 x16; V100s linked with NVLink
os kernel	Ubuntu20.04; kernel version not_extracted
driver runtime	Python3.10; CUDA11.6; PyTorch2.1.2; TVM0.14; NVIDIA driver not_extracted
models	ResNet152 319MB; DenseNet169 84MB; EfficientNetV2-s86MB; EfficientViT-b2 109MB; DistilBERT254MB; DistilGPT2 317MB; Swin-T training; section6.6 Llama2-13B26GB with Qwen2-0.5B training
tenant mix	six inference models duplicated round-robin to56 instances; co-located Swin-T training, batch72
batch load slo	Light/Heavy/Burst/Skewed and compressed Azure Functions traces; 300s runs; SLO=4x standalone inference execution; Tidle5s, watermark1GB
baseline	TaskSwitch implemented by authors, not vanilla PipeSwitch; SP-50/SP-75 static memory; UM+MPS; Infer-Only
latency	P99 request latency and request-level SLO compliance; LLM TTFT/TBT evaluated separately
throughput	training throughput in samples/s kept separate from inference metrics
resource usage	explicit dynamic shared GPU-memory pool; model sizes from Table3; no invented GPU utilization percentage
scalability	multi-GPU training evaluated separately in section6.2; excluded from local inference effect row
energy	not_extracted
reliability	Memory reclamation design is described; independent fault-injection result not_extracted

Field	Registered evidence
security fault isolation	resource handover and inference priority evaluated; no security/fault-containment guarantee inferred
overhead	memory adjustment/allocation and batch discard evaluated in section6.3; exact overhead values not_extracted
advantages	supports separable inference latency objectives while opportunistically admitting training
limitations	adapted baselines and model/training mix affect applicability; independent replication and repeated-run uncertainty remain open
tradeoffs	inference memory/SLO priority versus training throughput; hysteresis retains capacity to reduce thrashing

Reported outcomes

Outcome ID	GB-O035
Metric / value / unit	Average P99 inference latency improvement factor, as reported | 12.6 | x
Hardware / mechanism	single NVIDIA Tesla V100 16GB in section6.1; four-V100 host; separate section6.6 single A100 80GB LLM experiment | hybrid spatial sharing; CUDA VMM memory handover and dynamic SM masking; separate inference/training runtimes
Workload / conditions	section6.1 single-V100 DNN inference+Swin-T; Light/Heavy/Burst/Skewed/MAF traces;300s
Baseline	TaskSwitch (authors' adapted temporal-switch baseline)
Source anchor	printed p1666, section6.1/Figure10 discussion; PDF p11
Limit / interpretation	Author-reported average across tested workloads; not a uniform per-model speedup, and not a comparison to unmodified PipeSwitch.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

GB-P011 | ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

Field	Registered evidence
authors	Munkyu Lee; Sihoon Seong; Minki Kang; Jihyuk Lee; Gap-Joo Na; In-Geol Chun; Dimitrios Nikolopoulos; Cheol-Ho Hong
year	2024
venue	SC2024 identity; extracted text is arXiv2409.14447v1
doi	10.1109/SC41406.2024.00048
source	https://arxiv.org/pdf/2409.14447
source anchor	author preprint PDF pp4-5 Figures3-4; p8 experimental environment
mechanism	hybrid MIG+MPS; MPS processes within a MIG segment; profiler/configurator plus fleet allocator
gpu hardware	A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment
cpu memory	96vCPUs and1152GB main memory per cloud instance
os kernel	Ubuntu20.04; kernel not_extracted
driver runtime	CUDA12.0.1; PyTorch1.14.0 as printed in source (unverified version string); NVIDIA driver not_extracted
models	11 DNN workloads overall; local extracted example InceptionV3
tenant mix	one to three homogeneous MPS processes within a MIG instance; other models can occupy separate segments
batch load slo	extracted example:1-GPC MIG instance; batch4;1/2/3 MPS processes; no tail quantile identified for local latency
baseline	same InceptionV3/MIG/batch at1 MPS process versus2/3; fleet baselines not used for local effect
latency	reported latency in ms; percentile not identified for profiler point
throughput	requests/s; values354/444/446 at1/2/3 processes
resource usage	MIG GPC allocation; no GPU utilization percentage inferred from number of GPUs used
scalability	fleet allocator results remain separate from local measurement
energy	not_extracted
reliability	not_extracted
security fault isolation	MIG isolation is an architectural rationale; direct security/fault test not established
overhead	scheduling/profiling discussed; exact local overhead not_extracted

Field	Registered evidence
advantages	joint choice of MIG size, MPS process count and batch size exposes throughput/latency trade-off
limitations	preprint-version eligibility unresolved; some plotted instance sizes5/6 are interpolated, not real MIG profiles; local latency percentile unspecified
tradeoffs	additional MPS processes improve local throughput modestly while increasing latency under saturated small MIG instances

Reported outcomes

Outcome ID	GB-O036
Metric / value / unit	Reported throughput | 446 | requests/s
Hardware / mechanism	A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment | hybrid MIG+MPS; MPS processes within a MIG segment; profiler/configurator plus fleet allocator
Workload / conditions	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes
Baseline	same configuration with1 process (354requests/s)
Source anchor	author preprint PDF p5, sectionIII-B prose; Figure3 p4
Limit / interpretation	One local configuration, not fleet savings. Source is arXivv1; final published version not verified.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

Outcome ID	GB-O037
Metric / value / unit	Reported inference latency; quantile unspecified | 27 | ms
Hardware / mechanism	A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment | hybrid MIG+MPS; MPS processes within a MIG segment; profiler/configurator plus fleet allocator
Workload / conditions	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes
Baseline	same configuration with1 process (11ms)
Source anchor	author preprint PDF p5, sectionIII-B prose; Figure4 p4
Limit / interpretation	Quantile/arrival model not established for this profiler point; do not label p99. Publication-version check pending.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

GB-P012 | Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

Field	Registered evidence
authors	Minchen Yu; Ao Wang; Dong Chen; Haoxuan Yu; Xiaonan Luo; Zhuohao Li; Wei Wang; Ruichuan Chen; Dapeng Nie; Haoran Yang; Yu Ding
year	2025
venue	USENIX ATC2025 pp597-612
doi	not resolved
source	https://www.usenix.org/system/files/atc25-yu.pdf
source anchor	pp604-606 sections7.1-7.2, Table4, Figure8; PDF pp9-11
mechanism	API redirection/remoting plus temporal request-level sharing and host/GPU model swapping; GPU runtime sharing mode
gpu hardware	single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs
cpu memory	48vCPUs;384GB RAM per worker
os kernel	not reported in section7 settings
driver runtime	PyTorch-based clients replace CUDA libraries; exact OS/driver/CUDA versions not reported in section7 settings
models	ResNet152 local sharing test;8 models in Table4 including DenseNet169/201,InceptionV3,EfficientNet,ResNet50/101/152,Bert-qa
tenant mix	multiple low-frequency ResNet152 functions share one GPU; sufficient functions to saturate device memory
batch load slo	10-80 requests/minute/function; global defaultP98; CV deadline80ms and Bert-qa200ms
baseline	Native early binding; local remoting microbenchmark also compares GVirtuS; cluster INFless-KA results kept separate
latency	P98 under shared-GPU Figure8; Table4 isolated latency has no explicitly stated quantile
throughput	single-GPU throughput normalized to Native, request rates varied
resource usage	GPU load=time processing inference / observation time; host memory retains unloaded models
scalability	up to6 workers/4GPUs per worker evaluated separately; local outcome rows do not claim fleet effects
energy	not reported as measured outcome in checked sections7-8
reliability	not_extracted
security fault isolation	runtime sharing and runtime isolation modes differ; section7 measures sharing mode, not security guarantee

Field	Registered evidence
overhead	API redirection, swapping and extra CPU activity evaluated; ResNet152 remoting17ms versus Native25ms, Table4
advantages	more low-rate functions can share GPU memory capacity through late binding
limitations	PCIe swapping can hurt heavy models: Bert-qa144ms versus Native42ms in Table4; cluster data not a single-GPU speedup
tradeoffs	device-memory density versus transfer traffic, CPU work and tail latency

Reported outcomes

Outcome ID	GB-O038
Metric / value / unit	Reported single-GPU throughput factor | >10 | x
Hardware / mechanism	single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs | API redirection/remoting plus temporal request-level sharing and host/GPU model swapping; GPU runtime sharing mode
Workload / conditions	multiple ResNet152 functions on one V10032GB;10requests/minute/function
Baseline	Native early binding at same offered function rate
Source anchor	p606 section7.2; Figure8 p605; PDF pp10-11
Limit / interpretation	Lower bound reported by authors, not an exact10x; function count varies to saturate memory. No fleet cost claim.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

Outcome ID	GB-O039
Metric / value / unit	Reported P98 latency upper bound | <50 | ms
Hardware / mechanism	single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs | API redirection/remoting plus temporal request-level sharing and host/GPU model swapping; GPU runtime sharing mode
Workload / conditions	same single-GPU low-rate ResNet152 test,10-80requests/minute/function
Baseline	Native under matched request-rate setting
Source anchor	p606 section7.2; Figure8 p605; PDF pp10-11
Limit / interpretation	Bound from text, not a digitized point; default98th percentile from section7 metrics. Applies to this workload.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

GB-P013 | StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

Field	Registered evidence
authors	Hao Wu; Yue Yu; Junxiao Deng; Shadi Ibrahim; Song Wu; Hao Fan; Ziyue Cheng; Hai Jin
year	2024
venue	USENIX ATC2024 pp59-73
doi	not resolved
source	https://www.usenix.org/system/files/atc24-wu-hao.pdf
source anchor	pp67-70 sections9.1-9.4; Figures12-14,19; PDF pp10-13
mechanism	API forwarding into workflow-shared CUDA runtime/streams; PTB software spatial partitioning and fine-grained memory pool
gpu hardware	V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4
cpu memory	two Xeon Gold5117,28total cores,128GB DRAM
os kernel	not reported in section9 setup
driver runtime	PyTorch1.3.0,TensorFlow2.12,CUDA10.1 as reported; driver/OS version not reported in section9 setup
models	Image Processing,Traffic,Social Media workflows: SSD300,FSRCNN,ResNet152/50/18,VGG16,AlexNet,LSTM,BERT
tenant mix	concurrent functions within workflow share runtime; separate3/6-user Traffic test on one GPU
batch load slo	Azure Functions7-day invocation trace; sporadic/periodic/bursty; Traffic SLO200ms; up to3 workflows/runtime contexts per GPU in evaluated setup
baseline	INFless and Astraea with MPS; Stream-only; all warm-up/model-sharing/pipelined-loading settings disclosed in section9
latency	end-to-end CDF and SLO violations; no invented p99 scalar
throughput	average requests/s and within-experiment ratios
resource usage	runtime memory redundancy, automatic pool sizing and direct intra-GPU data sharing
scalability	single-GPU3/6-user limits versus separate4/8-node scaling tests
energy	not reported as measured outcome in checked section9
reliability	not_extracted
security fault isolation	functions within one workflow share CUDA address space; secure cross-workflow isolation is orthogonal; do not claim strong per-function isolation
overhead	API forwarding1-3microseconds;3-5microseconds at48functions; one-time profiling36s and transformation4s, p70

Field	Registered evidence
advantages	reduces redundant runtime memory and inter-function data copies
limitations	benefit diminishes across users; concurrency beyond3users caused queuing/SLO violations in evaluated setup
tradeoffs	workflow-local sharing and compiler/profiling preparation versus runtime density and startup latency

Reported outcomes

Outcome ID	GB-O040
Metric / value / unit	Author-reported average throughput factor | 6.7 | x
Hardware / mechanism	V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4 | API forwarding into workflow-shared CUDA runtime/streams; PTB software spatial partitioning and fine-grained memory pool
Workload / conditions	three workflow types,PyTorch/TensorFlow,Azure-derived invocation patterns; evaluated V100 setup
Baseline	INFless with documented warm-up/pipelined loading/model sharing
Source anchor	p68 section9.1/Figure13; PDF p11
Limit / interpretation	Aggregate within-paper result across tested workflows; not universal or a matched cross-paper speedup. Batch/concurrency optimizations contribute.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

Outcome ID	GB-O041
Metric / value / unit	Reported API-forwarding overhead range | 3-5 | microseconds
Hardware / mechanism	V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4 | API forwarding into workflow-shared CUDA runtime/streams; PTB software spatial partitioning and fine-grained memory pool
Workload / conditions	GPU API forwarding with48concurrent functions on one GPU
Baseline	direct API path; instrumentation overhead measurement
Source anchor	p70 section9.3/Figure18; PDF p13
Limit / interpretation	Per-API overhead, not end-to-end inference latency; many APIs asynchronous.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

GB-P014 | Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)

Field	Registered evidence
authors	Shulai Zhang; Ao Xu; Quan Chen; Han Zhao; Weihao Cui; Zhen Wang; Yan Li; Limin Xiao; Minyi Guo
year	2025
venue	USENIX ATC2025 pp1003-1019
doi	not resolved
source	https://www.usenix.org/system/files/atc25-zhang-shulai.pdf
source anchor	pp1011-1015 sections5.1,5.5-5.7,7; PDF pp10-14
mechanism	kernel-command-buffer interception; feedback-controlled temporal quotas and MIG spatial orchestration; Linux loadable module
gpu hardware	A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test
cpu memory	Xeon Silver4216,64cores as source reports; RAM not reported in Table1
os kernel	Ubuntu20.04; Linux5.15.0-91-generic
driver runtime	NVIDIA550.90.07;Docker24.0.7;CUDA12.1;Vulkan1.3;PyTorch2.2.1
models	local Transformer-XL inference batch8/16/32;ResNet50/MobileNetV2/BERT-Large also benchmarked
tenant mix	inference with co-running applications; separate training/rendering results not substituted
batch load slo	QoS:Poisson arrivals at half maximum throughput;50%temporal allocation;5/10/20/40/80ms tokens; target original soloP99
baseline	original soloP99 QoS target; fixed temporal quotas; adapted MIG baselines used separately for fleet packing
latency	fraction of requests exceeding originalP99 threshold
throughput	local normalized-throughput target errors separated from total GPU-count savings
resource usage	4.8MB host memory for kernel module;one CPU monitoring thread;NVML/DCGM monitoring
scalability	fleet packing evaluated separately; no local32.1%GPU-count claim
energy	not reported as measured outcome in checked section5
reliability	production reliability claimed in discussion; not independent fault-testing evidence
security fault isolation	separate virtual-device/application boundary; kernel module trust remains; security discussion is not proof of no vulnerabilities
overhead	token-length trade-off and4.8MBhost-memory module overhead

Field	Registered evidence
advantages	multi-runtime support and feedback-controlled quotas without application rewrites
limitations	MIG spatial partitions cannot be elastically reconfigured live; Linux kernel integration and trust required
tradeoffs	shorter scheduling tokens increase overhead; longer tokens increase waiting for larger batches

Reported outcomes

Outcome ID	GB-O042
Metric / value / unit	Reported average QoS violation rate | 1.23 | percent of requests
Hardware / mechanism	A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test | kernel-command-buffer interception; feedback-controlled temporal quotas and MIG spatial orchestration; Linux loadable module
Workload / conditions	Transformer-XL inference,batch8/16/32;50%GPUtime;Poisson arrivals;tokens5-80ms
Baseline	original soloP99 threshold (1%baseline violations by definition)
Source anchor	p1014 section5.5/Figure17; PDF p13
Limit / interpretation	Average across tested configurations; no claim that worst-case violation is1.23%. Threshold-relative metric, not latency in milliseconds.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

Outcome ID	GB-O043
Metric / value / unit	Reported kernel module host-memory footprint | 4.8 | MB
Hardware / mechanism	A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test | kernel-command-buffer interception; feedback-controlled temporal quotas and MIG spatial orchestration; Linux loadable module
Workload / conditions	kernel-space GPU sharing module on evaluated Linux host
Baseline	no module; additional host memory cost
Source anchor	p1014 section5.7; PDF p13
Limit / interpretation	Host memory, not GPU VRAM; separate monitoring thread also required.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

GB-P015 | XSched: Preemptive Scheduling for Diverse XPUs

Field	Registered evidence
authors	Weihang Shen; Mingcong Han; Jialong Liu; Rong Chen; Haibo Chen
year	2025
venue	USENIX OSDI2025 pp671-692
doi	not resolved
source	https://www.usenix.org/system/files/osdi25-shen-weihang.pdf
source anchor	p683 section7.4 case3/Figure15; PDF p14
mechanism	temporal/preemptive runtime interposition via XQueue/XShim; hardware-specific preemption levels
gpu hardware	single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes
cpu memory	not reported in extracted section7.4 case3
os kernel	not reported in extracted section7.4 case3
driver runtime	CUDA/Triton and Paella artifact; exact version numbers not reported in extracted case3
models	two Bert-large model clients for Triton; Paella mixed-model request workload for second case
tenant mix	one high-priority client at10requests/s;low-priority client continuous on sameGV100
batch load slo	Triton client rates as above;Paella workload lognormal sigma2.0;K-EDF K16; exact batch sizes not reported in extracted case3
baseline	vanilla Triton;Triton priority setting;standalone high-priority execution;separate comparison to Paella
latency	high-priorityP99 normalized to standalone;latency-throughput curve
throughput	requests/s in Paella curve;no aggregate cross-XPU overhead used as GPU inference result
resource usage	preemption enables foreground service priority; no invented GPU utilization percentage
scalability	ten-XPU portability evaluation distinguished from one-GPU case
energy	not reported as measured outcome in checked section7
reliability	not_extracted
security fault isolation	scheduling/performance priority, not a security or fault-containment guarantee
overhead	10lines Triton integration and15lines Paella-baseline integration;preemption overhead measured separately

Field	Registered evidence
advantages	adds priority policy to existing serving frameworks with small integration changes
limitations	preemption capability depends on vendor/hardware level;NPU latency gains do not transfer to GPU results
tradeoffs	foreground predictability versus background throughput and hardware-specific adaptation

Reported outcomes

Outcome ID	GB-O044
Metric / value / unit	Reported high-priorityP99 latency ratio | 1.07 | x standalone
Hardware / mechanism	single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes | temporal/preemptive runtime interposition via XQueue/XShim; hardware-specific preemption levels
Workload / conditions	twoBert-large clients ononeGV100;high-priority10req/s,backgroundcontinuous
Baseline	standalone high-priorityBert-large execution
Source anchor	p683 section7.4 case3/Figure15a; PDF p14
Limit / interpretation	Triton+XSched;not a rawmillisecond value. NativeTriton ratio1.53x;noNPUresultincluded.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

Outcome ID	GB-O045
Metric / value / unit	Author-reported P99 latency reduction | 30.0 | percent
Hardware / mechanism	single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes | temporal/preemptive runtime interposition via XQueue/XShim; hardware-specific preemption levels
Workload / conditions	same two-clientTriton/GV100case
Baseline	vanillaTriton (P99=1.53xstandalone)
Source anchor	p683 section7.4 case3/Figure15a; PDF p14
Limit / interpretation	Within-paper comparison only;author-rounded30.0%,not a universal bound.
Assessor / status	Codex-GroupB source assessment | AI_source_assessment;independent_pending

Calculated outcomes: formulas and inputs

Field	Calculation evidence
calculation_id	GB-CALC001
report_id	GB-P011
metric	Local throughput increase
formula	(446/354-1)*100
inputs	446requests/s at3processes;354requests/s at1process;InceptionV3,1-GPC,batch4
calculated_value	25.988700564971758
unit	percent
source_anchor	ParvaGPU arXivv1 PDF p5 sectionIII-B/Figure3
status	calculated,not reported;publication-version pending

Field	Calculation evidence
calculation_id	GB-CALC002
report_id	GB-P011
metric	Local latency factor
formula	27/11
inputs	27ms at3processes;11ms at1process;same1-GPC/batch4configuration
calculated_value	2.4545454545454546
unit	x
source_anchor	ParvaGPU arXivv1 PDF p5 sectionIII-B/Figure4
status	calculated,not reported;quantile unspecified;version pending